

Diversificacion Sectorial de Ecuador

HHI por ramas ISIC y ejercicios DEA de transicion productiva

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1 Resumen

Este reporte estudia la diversificacion sectorial de Ecuador con una base reproducible. El HHI se calcula con siete ramas ISIC mutuamente excluyentes a partir del Valor Agregado Bruto de la base UNSD. El analisis se complementa con un DEA CCR de frontera de produccion cuya unidad de decision es el par sector-ano y que modela trabajo y capital como insumos que generan valor agregado.

2 Datos

La tabla de inventario se genera desde el pipeline del proyecto.

	dataset	file	source	use
0	UNSD National Accounts Main Aggregates	outputs/tables/hhi_ecuador.csv	UNSD AMA API, GDP and breakdown at current prices	HHI de Ecuador y base para DEA por ano y por rama

	dataset	file	source	use
1	Ecuador sector comparable metrics	data/raw/ecuador/sector/BCECComparable_metrics_2018-2024p	WBICS Variables adicionales para enriquecer el DEA s...	
2	Ecuador sector capital context	data/raw/ecuador/sector/BCECapital_context_FBKFv_1965-2024p	Contexto de formacion bruta de capital fijo po...	
3	Ecuador sector exports status	data/raw/ecuador/sector/BCEExportsContext_estadoPrior de Bienes	Estado metodologico de exportaciones por rama ...	

3 Metodologia

3.1 HHI

Para cada ano:

$$HHI_t = \sum_i \left(\frac{X_{it}}{X_t} \right)^2$$

donde (X_{it}) es el valor de la rama (i) en el ano (t).

3.2 DEA CCR

El proyecto usa un modelo CCR orientado a insumos y reporta como control la regla de adecuacion:

$$n_{DMU} \geq 3(m + s)$$

donde (m) es el numero de inputs y (s) el numero de outputs.

El DEA se especifica como una frontera de produccion genuina: la unidad de decision es el par sector-ano y se modela trabajo (empleo) y capital (FBKF) como insumos independientes que producen valor agregado. Esta eleccion corrige dos problemas de versiones previas: (i) un modelo con un solo insumo y un solo producto colapsa a un ranking del cociente producto/insumo, y (ii) usar la productividad como producto introduce dependencia mecanica con el insumo, pues la productividad es exactamente ese cociente. Agrupar las siete ramas en la ventana comun 2018-2023 eleva el numero de DMU a 42, holgadamente por encima del umbral de adecuacion.

4 HHI de Ecuador

	year	hhi	classification
0	2000	0.1824	Moderately diversified

	year	hhi	classification
1	2001	0.1853	Moderately diversified
2	2002	0.1932	Moderately diversified
3	2003	0.2008	Moderately diversified
4	2004	0.2010	Moderately diversified
5	2005	0.1962	Moderately diversified
6	2006	0.1949	Moderately diversified
7	2007	0.1978	Moderately diversified
8	2008	0.1962	Moderately diversified
9	2009	0.2023	Moderately diversified
10	2010	0.2000	Moderately diversified
11	2011	0.1956	Moderately diversified
12	2012	0.1991	Moderately diversified
13	2013	0.1950	Moderately diversified
14	2014	0.1956	Moderately diversified
15	2015	0.2136	Moderately diversified
16	2016	0.2168	Moderately diversified
17	2017	0.2147	Moderately diversified
18	2018	0.2171	Moderately diversified
19	2019	0.2211	Moderately diversified
20	2020	0.2367	Moderately diversified
21	2021	0.2174	Moderately diversified
22	2022	0.2124	Moderately diversified
23	2023	0.2170	Moderately diversified
24	2024	0.2135	Moderately diversified

La particion de siete ramas evita el problema de pisos artificiales asociado a clasificaciones demasiado agregadas. La serie anual permite identificar cambios de concentracion sin imponer por construccion una clasificacion extrema.

	periodo	hhi_minimo	hhi_maximo	hhi_ultimo
0	2000-2024	0.1824	0.2367	0.2135

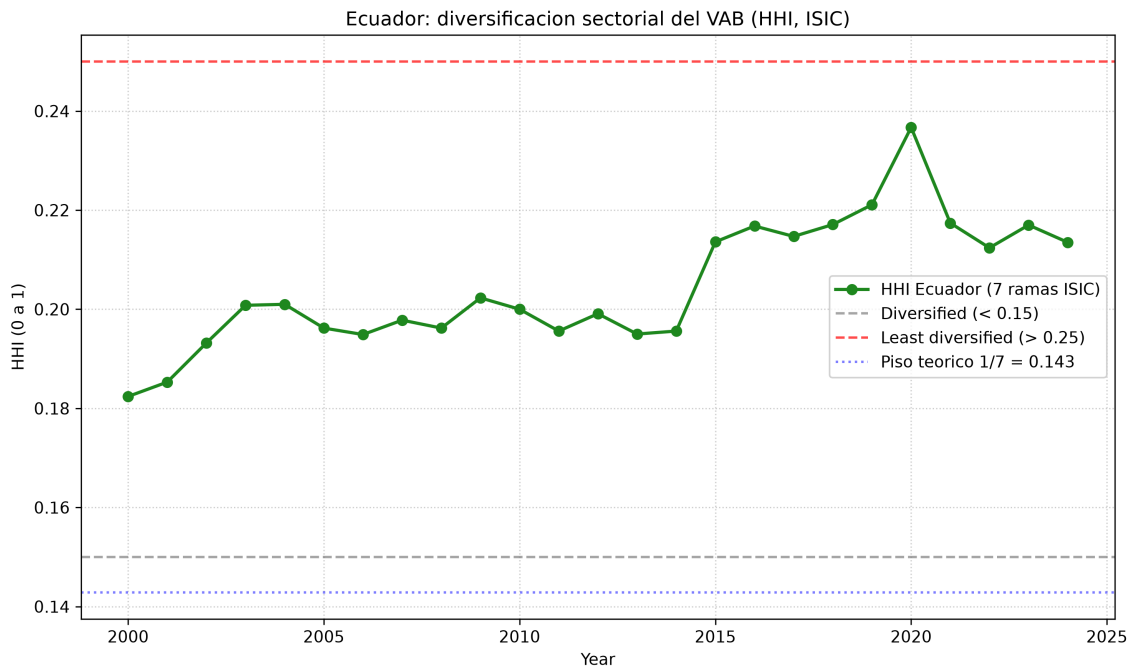


Figure 1: Ecuador HHI trend

5 DEA de Ecuador

model	dmu	inputs	outputs	n	threshold_3x	verificadict	efficiencia	inputs	outputs	used	age_note
Ecuador-Production_Frontier_DEA	2	input_employment	output_value_added	9	VALIDO	1	0.6519	input_employment	output_value_added		
	tor-year	input_capital	output_factor					input_capital	output_factor		
0											tor-año 2018-2023; tra-bajo (em-pleo) y...

5.1 DEA de frontera de produccion (sector-año)

dmu	factor	year	efficiency_ccr_input	status
Construction_2021	Construction	2021	1.000000	Optimization terminated successfully. (HiGHS S...
0				

	dmu	factor	year	efficiency_ccr_input	status
1	Construction_2023	Construction	2023	1.000000	Optimization terminated successfully. (HiGHS S...
2	Mining & utilities_2019	Mining & utilities	2019	1.000000	Optimization terminated successfully. (HiGHS S...
3	Mining & utilities_2022	Mining & utilities	2022	1.000000	Optimization terminated successfully. (HiGHS S...
4	Trade, restaurants & hotels_2020	Trade, restaurants & hotels	2020	0.964005	Optimization terminated successfully. (HiGHS S...
5	Construction_2019	Construction	2019	0.956741	Optimization terminated successfully. (HiGHS S...
6	Mining & utilities_2018	Mining & utilities	2018	0.946568	Optimization terminated successfully. (HiGHS S...
7	Mining & utilities_2021	Mining & utilities	2021	0.925893	Optimization terminated successfully. (HiGHS S...
8	Construction_2022	Construction	2022	0.908460	Optimization terminated successfully. (HiGHS S...
9	Mining & utilities_2023	Mining & utilities	2023	0.903027	Optimization terminated successfully. (HiGHS S...
10	Construction_2020	Construction	2020	0.896684	Optimization terminated successfully. (HiGHS S...
11	Construction_2018	Construction	2018	0.861992	Optimization terminated successfully. (HiGHS S...
12	Trade, restaurants & hotels_2021	Trade, restaurants & hotels	2021	0.839438	Optimization terminated successfully. (HiGHS S...

	dmu	factor	year	efficiency_ccr_input	status
13	Trade, restaurants & hotels_2022	Trade, restaurants & hotels	2022	0.775082	Optimization terminated successfully. (HiGHS S...
14	Trade, restaurants & hotels_2019	Trade, restaurants & hotels	2019	0.747117	Optimization terminated successfully. (HiGHS S...
15	Trade, restaurants & hotels_2023	Trade, restaurants & hotels	2023	0.740602	Optimization terminated successfully. (HiGHS S...
16	Trade, restaurants & hotels_2018	Trade, restaurants & hotels	2018	0.739670	Optimization terminated successfully. (HiGHS S...
17	Mining & utilities_2020	Mining & utilities	2020	0.704162	Optimization terminated successfully. (HiGHS S...
18	Other activities_2020	Other activities	2020	0.676285	Optimization terminated successfully. (HiGHS S...
19	Other activities_2023	Other activities	2023	0.606905	Optimization terminated successfully. (HiGHS S...
20	Transport & communication_2018	Transport & communication	2018	0.603651	Optimization terminated successfully. (HiGHS S...
21	Manufacturing_2018	Manufacturing	2018	0.599290	Optimization terminated successfully. (HiGHS S...
22	Other activities_2021	Other activities	2021	0.587998	Optimization terminated successfully. (HiGHS S...
23	Other activities_2022	Other activities	2022	0.585711	Optimization terminated successfully. (HiGHS S...
24	Transport & communication_2019	Transport & communication	2019	0.573775	Optimization terminated successfully. (HiGHS S...

	dmu	factor	year	efficiency_ccr_input	status
25	Other activities_2019	Other activities	2019	0.568126	Optimization terminated successfully. (HiGHS S...
26	Manufacturing_2021	Manufacturing	2021	0.567562	Optimization terminated successfully. (HiGHS S...
27	Manufacturing_2023	Manufacturing	2023	0.560527	Optimization terminated successfully. (HiGHS S...
28	Manufacturing_2022	Manufacturing	2022	0.552537	Optimization terminated successfully. (HiGHS S...
29	Manufacturing_2020	Manufacturing	2020	0.549259	Optimization terminated successfully. (HiGHS S...
30	Transport & communication_2020	Transport & communication	2020	0.539946	Optimization terminated successfully. (HiGHS S...
31	Other activities_2018	Other activities	2018	0.520814	Optimization terminated successfully. (HiGHS S...
32	Manufacturing_2019	Manufacturing	2019	0.519486	Optimization terminated successfully. (HiGHS S...
33	Transport & communication_2021	Transport & communication	2021	0.493122	Optimization terminated successfully. (HiGHS S...
34	Transport & communication_2023	Transport & communication	2023	0.439192	Optimization terminated successfully. (HiGHS S...
35	Transport & communication_2022	Transport & communication	2022	0.394357	Optimization terminated successfully. (HiGHS S...
36	Agriculture, forestry & fishing_2023	Agriculture, forestry & fishing	2023	0.274628	Optimization terminated successfully. (HiGHS S...

	dmu	factor	year	efficiency_ccr_input	status
37	Agriculture, forestry & fishing_2018	Agriculture, forestry & fishing	2018	0.258894	Optimization terminated successfully. (HiGHS S...
38	Agriculture, forestry & fishing_2021	Agriculture, forestry & fishing	2021	0.254435	Optimization terminated successfully. (HiGHS S...
39	Agriculture, forestry & fishing_2020	Agriculture, forestry & fishing	2020	0.249551	Optimization terminated successfully. (HiGHS S...
40	Agriculture, forestry & fishing_2019	Agriculture, forestry & fishing	2019	0.249393	Optimization terminated successfully. (HiGHS S...
41	Agriculture, forestry & fishing_2022	Agriculture, forestry & fishing	2022	0.244916	Optimization terminated successfully. (HiGHS S...

Cada DMU es un par sector-año. Los insumos son empleo (trabajo) y FBKF (capital), dos factores de producción independientes provenientes del BCE (MEI 2018-2024 y FBKF 1965-2024), y el producto es el valor agregado por rama. La eficiencia mide que tan cerca está cada sector-año de la frontera definida por las combinaciones observadas de trabajo y capital que generan más valor agregado.

5.1.a Eficiencia media por rama

	factor	mean_efficiency_sector
0	Construction	0.937313
1	Mining & utilities	0.913275
2	Trade, restaurants & hotels	0.800986
3	Other activities	0.590973
4	Manufacturing	0.558110
5	Transport & communication	0.507340
6	Agriculture, forestry & fishing	0.255303

Agregar los años por rama ofrece una lectura sectorial de la eficiencia relativa sobre la ventana común 2018-2023.

5.1.b Inferencia bootstrap (Simar-Wilson)

Las eficiencias DEA son estimadores deterministas, sesgados hacia arriba y sin error estándar: la frontera estimada queda por dentro de la verdadera. Antes de cualquier lectura inferencial

corregimos ese sesgo y construimos intervalos de confianza con el bootstrap suavizado de Simar & Wilson (1998), con reflexion en la frontera y correccion de media y varianza.

	dmu	factor	efficiency	CCR_in	CCR_out	bias_corrected	ci_low	ci_high
0	Construction_2023	Construction	2023	1.0000	0.1406	0.8594	0.7539	0.9181
1	Construction_2021	Construction	2021	1.0000	0.1814	0.8186	0.6593	0.9093
2	Construction_2019	Construction	2019	0.9567	0.1438	0.8129	0.6922	0.8794
3	Trade, restaurants & hotels_2020	Trade, restaurants & hotels	2020	0.9640	0.1559	0.8081	0.6456	0.8898
4	Construction_2022	Construction	2022	0.9085	0.1230	0.7855	0.6935	0.8382
5	Mining & utilities_2019	Mining & utilities	2019	1.0000	0.2274	0.7726	0.4234	0.9099
6	Construction_2018	Construction	2018	0.8620	0.1088	0.7532	0.6775	0.7965
7	Construction_2020	Construction	2020	0.8967	0.1524	0.7443	0.6086	0.8204
8	Mining & utilities_2018	Mining & utilities	2018	0.9466	0.2036	0.7430	0.4221	0.8662
9	Mining & utilities_2022	Mining & utilities	2022	1.0000	0.2896	0.7104	0.3291	0.9096
10	Trade, restaurants & hotels_2021	Trade, restaurants & hotels	2021	0.8394	0.1406	0.6988	0.5620	0.7733
11	Mining & utilities_2021	Mining & utilities	2021	0.9259	0.2389	0.6870	0.3517	0.8483
12	Trade, restaurants & hotels_2022	Trade, restaurants & hotels	2022	0.7751	0.1170	0.6581	0.5567	0.7122
13	Mining & utilities_2023	Mining & utilities	2023	0.9030	0.2515	0.6515	0.3073	0.8300

	dmu	factor	efficiency	CCR_in	CCR_out	efficiency_bias_corrected	ci_low	ci_high
14	Trade, restaurants & hotels_2023	Trade, restaurants & hotels	2023	0.7406	0.1036	0.6370	0.5587	0.6806
15	Trade, restaurants & hotels_2018	Trade, restaurants & hotels	2018	0.7397	0.1235	0.6161	0.4948	0.6818
16	Trade, restaurants & hotels_2019	Trade, restaurants & hotels	2019	0.7471	0.1321	0.6150	0.4961	0.6846
17	Other activities_2020	Other activities	2020	0.6763	0.0835	0.5928	0.5236	0.6263
18	Mining & utilities_2020	Mining & utilities	2020	0.7042	0.1548	0.5493	0.3221	0.6432
19	Transport & communication_2018	Transport & communication	2018	0.6037	0.0682	0.5354	0.4894	0.5594
20	Manufacturing_2018	Manufacturing	2018	0.5993	0.0685	0.5308	0.4867	0.5551
21	Other activities_2023	Other activities	2023	0.6069	0.0859	0.5210	0.4339	0.5611
22	Transport & communication_2019	Transport & communication	2019	0.5738	0.0655	0.5082	0.4632	0.5316
23	Other activities_2021	Other activities	2021	0.5880	0.0814	0.5066	0.4254	0.5440
24	Other activities_2022	Other activities	2022	0.5857	0.0824	0.5033	0.4197	0.5417
25	Manufacturing_2021	Manufacturing	2021	0.5676	0.0648	0.5028	0.4608	0.5258
26	Manufacturing_2023	Manufacturing	2023	0.5605	0.0633	0.4972	0.4544	0.5194
27	Manufacturing_2022	Manufacturing	2022	0.5525	0.0624	0.4901	0.4480	0.5120
28	Other activities_2019	Other activities	2019	0.5681	0.0798	0.4883	0.4076	0.5254

	dmu	factor	efficiency	CCR_in	CCR_out	efficiency_bias_corrected	ci_low	ci_high
29	Manufacturing_2020	Manufacturing	2020	0.5493	0.0638	0.4855	0.4448	0.5084
30	Transport & communication_2020	Transport & communication	2020	0.5399	0.0616	0.4784	0.4363	0.5003
31	Manufacturing_2019	Manufacturing	2019	0.5195	0.0589	0.4606	0.4214	0.4814
32	Other activities_2018	Other activities	2018	0.5208	0.0776	0.4432	0.3668	0.4800
33	Transport & communication_2021	Transport & communication	2021	0.4931	0.0579	0.4352	0.3935	0.4564
34	Transport & communication_2023	Transport & communication	2023	0.4392	0.0552	0.3840	0.3366	0.4066
35	Transport & communication_2022	Transport & communication	2022	0.3944	0.0519	0.3425	0.2948	0.3652
36	Agri-culture, forestry & fishing_2023	Agri-culture, forestry & fishing	2023	0.2746	0.0374	0.2372	0.2093	0.2532
37	Agri-culture, forestry & fishing_2018	Agri-culture, forestry & fishing	2018	0.2589	0.0361	0.2228	0.1955	0.2380
38	Agri-culture, forestry & fishing_2021	Agri-culture, forestry & fishing	2021	0.2544	0.0355	0.2189	0.1915	0.2339
39	Agri-culture, forestry & fishing_2019	Agri-culture, forestry & fishing	2019	0.2494	0.0355	0.2139	0.1873	0.2289
40	Agri-culture, forestry & fishing_2022	Agri-culture, forestry & fishing	2022	0.2449	0.0324	0.2125	0.1885	0.2262

	dmu	factor efficiency	year	CCR_in	CCR_out	efficiency_bias_corrected	ci_low	ci_high
		forestry & fishing						
41	Agri-culture, forestry & fishing	Agri-culture, forestry & fishing	2020	0.2496	0.0406	0.2090	0.1730	0.2289

La columna `efficiency_bias_corrected` es la eficiencia corregida por sesgo y `[ci_low, ci_high]` su intervalo de confianza al 95%. Que las DMU con eficiencia DEA igual a 1 reciban una correccion estrictamente menor que 1 confirma el sesgo al alza esperado. Solo deben interpretarse como diferencias robustas de eficiencia aquellas cuyas bandas no se solapan.

5.2 Contexto de capital

	factor	coverage_start_year	coverage_end_year	fbkf_avg_musd	fbkf_latest_musd	source	notes
0	Agri-culture, forestry & fishing	2018	2023	2.547096e+06	2.804245e+06	BCE FBKF 1963-2024p: https://contenido.bce.fin...	promedio y ultimo dato 2018-2023 por rama...
1	Construc-tion	2018	2023	3.916281e+05	4.139441e+05	BCE FBKF 1963-2024p: https://contenido.bce.fin...	promedio y ultimo dato 2018-2023 por rama...
2	Manufac-turing	2018	2023	2.897816e+06	3.398009e+06	BCE FBKF 1963-2024p: https://contenido.bce.fin...	promedio y ultimo dato 2018-2023 por rama...
3	Mining & utilities	2018	2023	2.531196e+06	2.337590e+06	BCE FBKF 1963-2024p: https://contenido.bce.fin...	promedio y ultimo dato

id	category	coverage_start_year	coverage_end_year	fbkf_avg_musd	fbkf_latest_musd	source	notes
							2018-2023 por rama...
4	Other activities	2018	2023	9.816578e+06	1.042957e+07	BCE FBKF 1963-2024p: https:// contenido. bce.fin...	promedio y ultimo dato 2018-2023 por rama...
5	Trade, restaurants & hotels	2018	2023	1.533717e+06	2.122502e+06	BCE FBKF 1963-2024p: https:// contenido. bce.fin...	promedio y ultimo dato 2018-2023 por rama...
6	Transport & communication	2018	2023	2.040258e+06	2.499636e+06	BCE FBKF 1963-2024p: https:// contenido. bce.fin...	promedio y ultimo dato 2018-2023 por rama...

Esta tabla resume la FBKF promedio por rama armonizada. La FBKF entra al DEA de frontera de produccion como insumo de capital a nivel sector-ano; aqui se muestra ademas su promedio por rama como contexto descriptivo. En la salida actual, la cobertura efectiva de la FBKF armonizada llega hasta 2023, lo que define la ventana comun 2018-2023 del panel DEA.

5.3 Estado de exportaciones

id	status	coverage_start_year	coverage_end_year	source	use	reason
0	PENDIENTE_VINCULAR	NaN	NaN	BCE Comercio Exterior de Bienes	Segundo output del DEA de frontera (output exp...)	La evidencia oficial de localizada en BCE esta pu...

Las exportaciones aun no entran al DEA porque la evidencia oficial del BCE esta organizada por productos y por paises, no por una clasificacion de ramas equivalente a la

particion ISIC de siete sectores. El pipeline ya esta preparado para incorporarlas como segundo producto: si se provee una serie verificada por rama en `data/raw/ecuador/sector_exports_by_branch.csv` (`factor`, `year`, `exports`) que cubra las siete ramas en la ventana del panel, `build_ecuador_sector_metrics.py` la agrega como `output_exports` y el DEA recalcula la regla de adecuacion (con 42 DMU admite hasta cuatro variables manteniendo el veredicto VALIDO). Mientras no exista ese correlacionador oficial, la compuerta permanece cerrada y el estado queda como `PENDIENTE_VERIFICAR`.

6 Discusion

El HHI ofrece una lectura descriptiva mas estable cuando la particion sectorial es suficientemente fina. El DEA de frontera de produccion debe interpretarse con cautela: identifica posiciones relativas de eficiencia tecnica bajo rendimientos constantes a escala, no relaciones causales. Con 42 DMU el modelo satisface la regla de adecuacion, y la inferencia por bootstrap (Simar-Wilson) corrige el sesgo al alza y delimita intervalos de confianza, de modo que las comparaciones entre ramas se sostienen solo cuando sus bandas no se solapan.

7 Siguiete paso

El siguiente paso es completar y verificar una base sectorial armonizada en `data/raw/ecuador/sector_comparable_metrics.csv` para incorporar empleo, productividad, exportaciones o formacion de capital con cobertura temporal y sectorial consistente. Hasta cerrar esa evidencia, cualquier fuente adicional debe tratarse como `PENDIENTE_VERIFICAR`.